

Four permutation-fair dice with a three-face die

Exact construction and certificate, 30 September 2026

Status. Verified twice by independent exact-arithmetic programs, including direct integer sub-sequence counting. This is an existence result, not an optimization of the other three dice.

Theorem 1. *There are four permutation-fair uniform dice with face counts*

$$\begin{aligned} K_A &= 163693699141790847787594200000, \\ K_B &= 3055345920, \quad K_C = 5508, \quad K_D = 3. \end{aligned}$$

Consequently, the minimum possible size of a prescribed die in a four-die permutation-fair set is exactly three, using the universal $n-1$ lower bound established in the companion monotone-tensor note.

The exact certificate. The table below specifies a run-length word. A row labelled (i, k) means k consecutive occurrences of die i ; the rows are concatenated in order. Number all letters in this expanded word consecutively, and assign each number to the indicated die. All labels are distinct. There are exactly three occurrences of D .

For verification without expanding the very large runs, start with $C_\emptyset = 1$ and all other partial-pattern counts zero. Reading a run i^k , simultaneously update

$$C_{\pi i} \leftarrow C_{\pi i} + kC_\pi \quad (i \notin \pi).$$

No pattern has a repeated letter, so it can select at most one occurrence from the current run; this recurrence is exact. Applied to the table, it gives for every $\pi \in S_4$

$$C_\pi = \frac{K_A K_B K_C K_D}{24}.$$

The independent program `verify_simple_three_face_construction.py` checks this identity by integer arithmetic, and separately checks all weighted order probabilities by rational arithmetic. The machine-readable run table is `four_dice_three_face_simple_integer.json`.

How the example was found. At the three faces of D , the cumulative probabilities of the old dice are

	D face 1	D face 2	D face 3
A	1/7	71/140	17/20
B	5/33	47/96	907/1056
C	133/918	77/153	23/27.

Their centered versions $y_j = 2p_j - 1$ satisfy

$$\sum_q y_j(q) = 0, \quad \sum_q y_i(q)y_j(q) = 1 \ (i \neq j), \quad \sum_q y_A(q)y_B(q)y_C(q) = 0.$$

These identities give each subset of dice the probability prescribed by a uniform ranking of being exactly the set below D . They alone do not ensure the full order law. Four interval blocks were then filled with positive rational weighted dice. A linear program solved the remaining

order constraints; rational elimination converted its solution into an exact certificate before any assertion of fairness.

The construction uses the classical three-die fair word

abccbba bcaacb cabbac

as a local starting point, recorded in Purcell, *Building Sets of Permutation-Fair Dice*, Example 5, doi:10.1080/00029890.2024.2345576. The four rational density tilts used for B are

$$-\frac{121}{95}, \quad \frac{39391}{83895}, \quad -\frac{319}{7020}, \quad -\frac{2}{5}.$$

Die C has six equal-weight atoms within each interval. The weights of A are the exact solution of the final linear system. Clearing the three old dice's denominators gives the integer word below and leaves D with three faces.

Run	Die	Length
1	A	1457525804001828318048582655
2	B	28019520
3	A	5891126255200470770758794782
4	C	133
5	A	312233865893543192502225816
6	C	133
7	A	777545070923506526991072450
8	B	60776640
9	A	783148640385362733842852349
10	B	60776640
11	A	777545070923506526991072450
12	C	133
13	A	917140392154551310254949284
14	C	133
15	A	777545070923506526991072450
16	B	93533760
17	A	777545070923506526991072450
18	C	133
19	A	3151103708479473819911188350
20	B	109912320
21	A	777545070923506526991072450
22	B	109912320
23	A	777545070923506526991072450
24	C	133
25	A	6207265071456708948105572064
26	D	1
27	A	706544614483005757601374890
28	B	212567706
29	A	10788535011066461264007101442
30	C	329
31	A	895698472294965317331540681
32	C	329
33	A	777545070923506526991072450
34	B	185624262
35	A	5016949959657333152877376734
36	B	185624262
37	A	5043402870558576020335777302
38	C	329
39	A	738653884932274504567638147
40	C	329
41	A	31776220877404439372527786104

Run	Die	Length
42	B	158680818
43	A	777545070923506526991072450
44	C	329
45	A	777545070923506526991072450
46	B	145209096
47	A	777545070923506526991072450
48	B	145209096
49	A	777545070923506526991072450
50	C	329
51	A	777545070923506526991072450
52	D	1
53	A	3974274190980916216428391812
54	B	183792795
55	A	781729313513001301148880816
56	C	320
57	A	732178531447067349175653486
58	C	320
59	A	17110902371291397319237221726
60	B	186641465
61	A	777545070923506526991072450
62	C	320
63	A	777545070923506526991072450
64	B	188065800
65	A	777545070923506526991072450
66	B	188065800
67	A	8705230920360437285344259556
68	C	320
69	A	777545070923506526991072450
70	C	320
71	A	777545070923506526991072450
72	B	190914470
73	A	777545070923506526991072450
74	B	190914470
75	A	19376423167413782652617525454
76	C	320
77	A	777545070923506526991072450
78	D	1
79	A	770997322957834893079568682
80	B	57480624
81	A	777545070923506526991072450
82	C	136
83	A	777545070923506526991072450
84	C	136
85	A	13832117577481326638051709900
86	B	67060728
87	A	777545070923506526991072450
88	B	67060728
89	A	777545070923506526991072450
90	C	136
91	A	777545070923506526991072450
92	C	136
93	A	777545070923506526991072450
94	B	76640832
95	A	777545070923506526991072450
96	C	136
97	A	777545070923506526991072450
98	B	81430884

Run	Die	Length
99	A	777545070923506526991072450
100	B	81430884
101	A	2175489261594400367097126918
102	C	136
103	A	777545070923506526991072450